

selective pressures

evolutionary inertia

leaf expansion during fowering

0% no leaf development

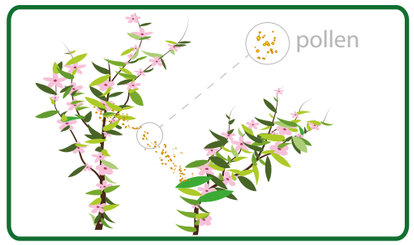
100% full canopy

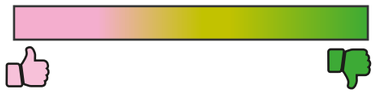
a. wind pollination

Hypothesis: Leafess fowering reduces barriers to pollen movement, allowing for more efcient pollen transfer.

Prediction: Strong association between FLS and pollination syndrome.



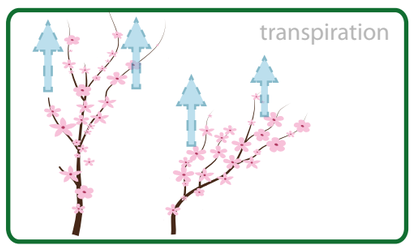


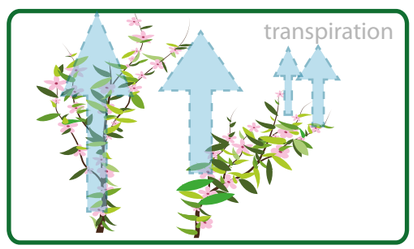


b. water dynamics

Hypothesis: Leafess fowering reduces hydraulic demand by temporally separating foral hydration and leaf transpiration.

Prediction: Strong association between FLS and hydraulic demand.







c. early fowering

Hypothesis: Leafess fowering is a byproduct of selection for early fowering and has no inherent function.

Prediction: Strong association between FLS and fowering time.







d. phylogenetic conservatism vs phylogenetic lability

Hypothesis: Leafess fowering is phylogenetically conserved and FLS variation refects macro-evolutionary relationships among taxa

Prediction: Strong phylogenetic signal in FLS variation.

